

Technical & Commercial Proposal For BWRO Plant

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Date: Oct 20, 2026 Ref: PS/PRO/26/0646

TECHNICAL PROPOSAL:
10,000 GPD BWRO PLANT (SEMI-AUTOMATIC)

1. DESIGN PARAMETERS

Raw water/Feed Flow	10,000 GPD
System Recovery	50%
Product Flow	5,000 GPD
Reject Flow	5,000 GPD
Design TDS	5,000 ppm
Product TDS	<300 ppm
RO Inlet Pressure	15~20 Bar(g)
Design Temperature	25°C

2. MULTIMEDIA FILTER VESSEL

Brand	Imported.
Material	FRP
Size	16" x 65"
Flow rate	03~05 m^3/hr.
Operating Temp.	49°C
Operating Pressure	3~4 Bar(g)
Filling Media	Gravel, Sand & A. Carbon
Quantity	01

3. CARTRIDGE FILTER HOUSING

Model	PVC Housing for Slim Filters
Origin	Imported
Material of Construction	PVC
Size	20" Jumbo
Design Pressure	3~4 Bar(g)
No. of Filters	05 Filters/Housing
Location	Feed Water Line
Quantity	01

4. CARTRIDGE FILTER

Make/Made	Hydro-Safe
Application	Filtration
Material of Construction	100% PPF
ID / OD	25mm / 65mm
Length of Filter	20" Jumbo
Filtration Frequency	05 micron
Chemical Resistant	Excellent
Max. Temperature	45°C
Quantity	01

5. DOSING PUMP

Model	PKX MA/A
Make	ETATRON
Origin	Italy
Application	Chemical Dosing
Max. Flow Rate	05 Lit/Hr.
Pressure	05 Bar(g)
Housing	PP Reinforced
Pump Head	PP with Lip or Ceramic Ball Valve
Diaphragm	PTFE
Stroke Rate	120 impulses/min
Valve	FPM (Viton)
Injection Volume	0.63 cm3
Max. Fluid Temperature	45°C
Duty Cycle	Continue
Connection (ID × OD)	4mm x 6mm
Quantity	01

6. DOSING TANK

Brand/Manufacturer	Local
Material of Construction	HDPE/PP
Capacity	80 Liters
Quantity	01

7. CLEANING TANK

Brand/Manufacturer	Local
Material of Construction	HDPE/PP
Capacity	80 Liters
Quantity	01

8. FEED PUMP

M	
Make	Grundfos
Rated Flow	02 m3/hr.
Rated Head	3~4 Bar(g)
MOC	SS 316
Motor Type	IEC – 132SC
Power Req.	1.1 kW
Frequency	50 Hz
Voltage	3 x 380-415D V
Rated Speed	2910 – 2930 rpm
Insulation Class	F
Protection	IP55
Quantity	01

9. HP PUMP

Model	CRN / CNP
Make	Grundfos
Rated Flow	02 m3/hr.
Rated Head	15~20 Bar(g)
MOC	SS 316
Motor Type	IEC – 132SC
Power Req.	5.5 kW
Frequency	50 Hz
Voltage	3 x 200D V
Rated Speed	2920 – 2940 rpm
Insulation Class	F
Protection	IP55
Quantity	01

10. MEMBRANE HOUSING

Brand/Manufacturer	China
Origin	Imported
Material of Construction	FRP
Max. Operating Pressure	300 psi
Model	Single-Element Housing
Quantity	01

11. RO MEMBRANE

Model	BW30 PRO-400
Make	Dupont Filmtec OR Toray
Origin	USA
Specs	Polyamide Thin-Film Composite
Dia	8 inches
Length	40 inches
Active Area	400 ft2 (37m2)
Permeate Flow Rate	7,500 GPD (28m3/d)
Min Salt Rejection	99.6%
Max Salt Rejection	99.8%
Max Operating Temp.	113°F (45°C)
Quantity	01

12. INSTRUMENTATION:

TDS CONTROLLER

Brand	ROC or Equivalent
Origin	China
Type	Digital
Location	Feed & Product Line
Quantity	01
SENSOR	
TDS Range	5,000 ppm
Cable Length	6m
Quantity	01

FLOW METER

Brand	LZ Series or Equivalent
Origin	Imported
Type	Rota-Type
Range	25 GPM
DN	50
Accuracy	± 0.10%
Operating Temperature	60°C
Max Pressure	06 Bar (g)
Material of Construction	PE Body
Seal	O-Ring (Viton)
Location	Product & Concentration Line
Quantity	02

PRESSURE INDICATOR

Brand	Wika or Equivalent
Origin	Imported
Range	0~6 & 0~25 Bar(g)
Dial	Liquid Filled
Location	Field Mounted – Feed & Concentrated Line
Material of Construction	SS
Ambient Temperature	68°F
Quantity	As Per Requirement

13. CONTROL PENAL

BREAKERS, OVERLAOD RELAYS, TIMER, CABLES, ETC.

Brand/Manufacturer	Any Foreign
Origin	France / Taiwan / China

CONTROL TYPE

Mode	Semi-Automatic

PENAL BOX

Material of Construction	MS – Powder Coated
Standard	NEMA–4 Enclosure
Origin	Local Fabricated
Quantity	01

14. PIPING & FITTINGS

LOW & HIGH-PRESSURE PIPES, FITTINGS & VALVES

Material of Construction	PVS Sch#80
Brand/Manufacturer	Hydro-Plast / Hydro-Seal / Equivalent
Origin	Imported
Quantity	01 Lot

SAMPLING VALVE

Application	¼ Quick Connection
Brand/Manufacturer	Any Foreign
Material of Construction	PVDF / HEDP
Quantity	As Per Requirement

15. SKID

Material of Construction	SS 202 – Epoxy Coated
Quantity	01

COMMERCIAL PROPOSAL:

10,000 GPD BWRO PLANT. (SEMI-AUTOMATIC)

The cost of complete RO Plant system installation with components and equipment mentioned in Technical Proposal & commissioning is as given below;

PRICE SUMMARY

COMPLETE RO PLANT 10,000 GPD BWRO.	PKR 735,000.00

PAYMENT TERMS & CONDITIONS

- **75% Advance** for RO System at the time of Purchase Order.
- **15%** after delivery.
- **10%** after Installation and Commissioning.
- All payments should be released within **10 days** from the date of invoice submission.

DELIVERY

- Within 02~04 weeks after receiving confirmed **Purchase Order & Advance.**

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